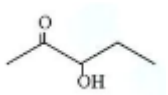
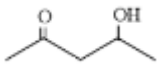
Quizrr Chapter-Wise Test for JEE Advanced - 2025

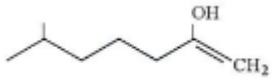
By: C I P H Ξ R

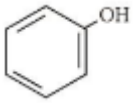
Q.1

Which is more easily dehydrated?

(A) 

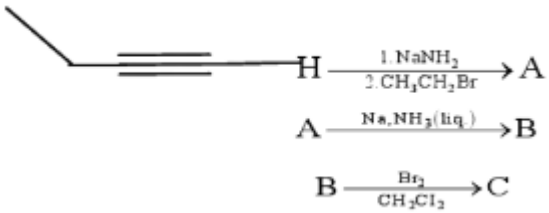
(B) 

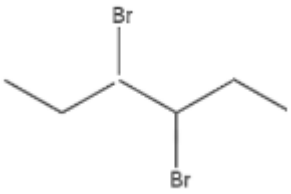
(C) 

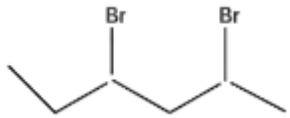
(D) 

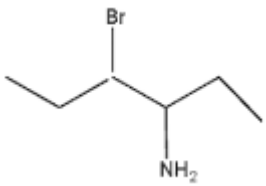
Q.2

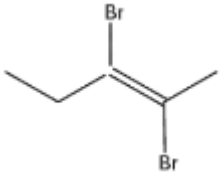
What is the final product, C, of the following reaction sequence?



(A) 

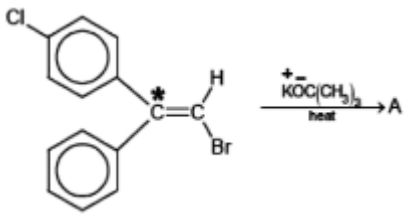
(B) 

(C) 

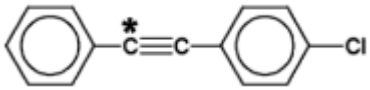
(D) 

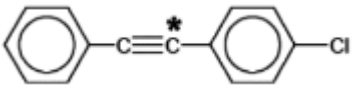
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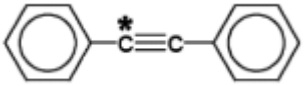
Q.3

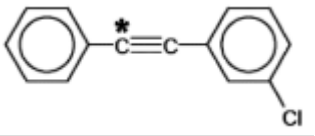


A is?

(A) 

(B) 

(C) 

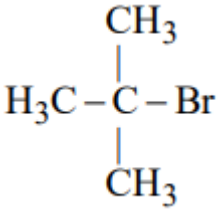
(D) 

Q.4

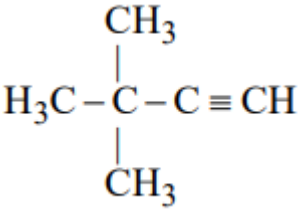
Outline the best synthesis of 4,4-dimethyl, 2-pentyne using $NaNH_2$ in liquid NH_3 using any two of these -

(i) $HC \equiv CH$

(ii)



(iii)



(iv) CH_3I

(v) $CH_3 - C \equiv CBr$

(A) (i) + (ii)

(B) (iii) + (v)

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(C) (iii) + (iv)

(D) (ii) $+(\mathrm{v})$

Q.5

A hydrocarbon P having molecular formula $\mathrm{C_8H_{10}}$ can form only three monochloroderivative. P on oxidation with $\mathrm{KMnO_4}$ yields a compound, which on heating with aq. $\mathrm{NH_3}$ yields Q. Q on treatment with NaOH followed by NaOCl yields a salt which on acidification yields a compound R having molecular formula $\mathrm{C_7H_7O_2\ N}$. The compound R is further treated with $(\mathrm{NaNO_2 + HCl})$ and then with base to give dipolar ion S which readily losses two gases to give an intermediate T is finally converted into hydrocarbon U on dimerisation. Degree of unsaturation in the hydrocarbon U will be

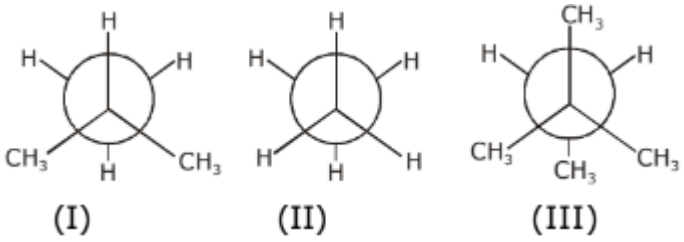
(A) 1

(B) 2

(C) 6

(D) 9

Q.6



Correct order of mono bromination of given compound is :

(A) I > III > II

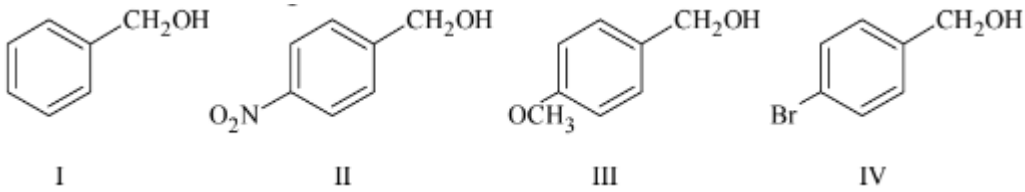
(B) I > II > III

(C) II > I > III

(D) III > II > I

Q.7

Consider the following alcohols:



The order of decreasing reactivities of these alcohols towards substitution with HBr is:

(A) III > I > IV > II

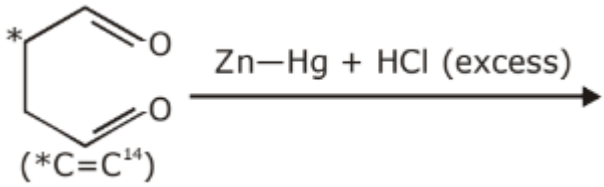
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(B) III > I > II > IV

(C) I > III > IV > II

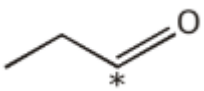
(D) I > III > II > IV

Q.8

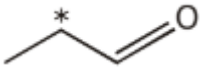


product will be :

(A) 

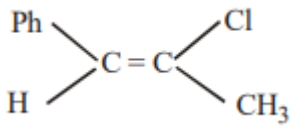
(B) 

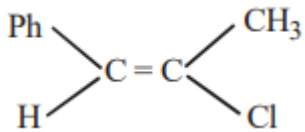
(C) 

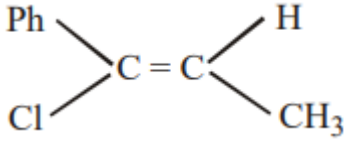
(D) 

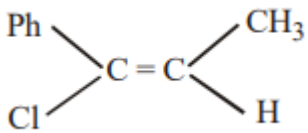
Q.9

Predict the major product(s) of following reaction $\mathrm{Ph-C\equiv C-CH_3 + 1\ equivalent\ of\ HCl \longrightarrow}$

(A) 

(B) 

(C) 

(D) 

Q.10

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Trans 2-butene on careful reaction with bayer reagent yields racemic $(d+\ell)$ mixture product this is example of -

(A) Stereospecific reaction

(B) Stereoselective reaction

(C) Stereoequilibrium reaction

(D) Syn addition

Q.11

Which of the following are correctly matched

(A) Hydrolysis of phenyl magnesium iodide - Benzene

(B) γ - Isomer of BHC - Lindanne

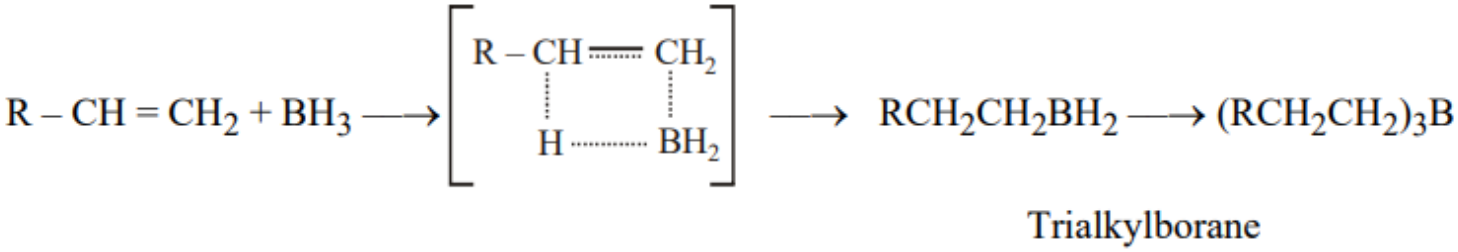
(C) $(2n+4)\pi$ rule $-$ Aromaticity

(D) Trimerisation of propyne - Mesitylene

Q.12

Passage:

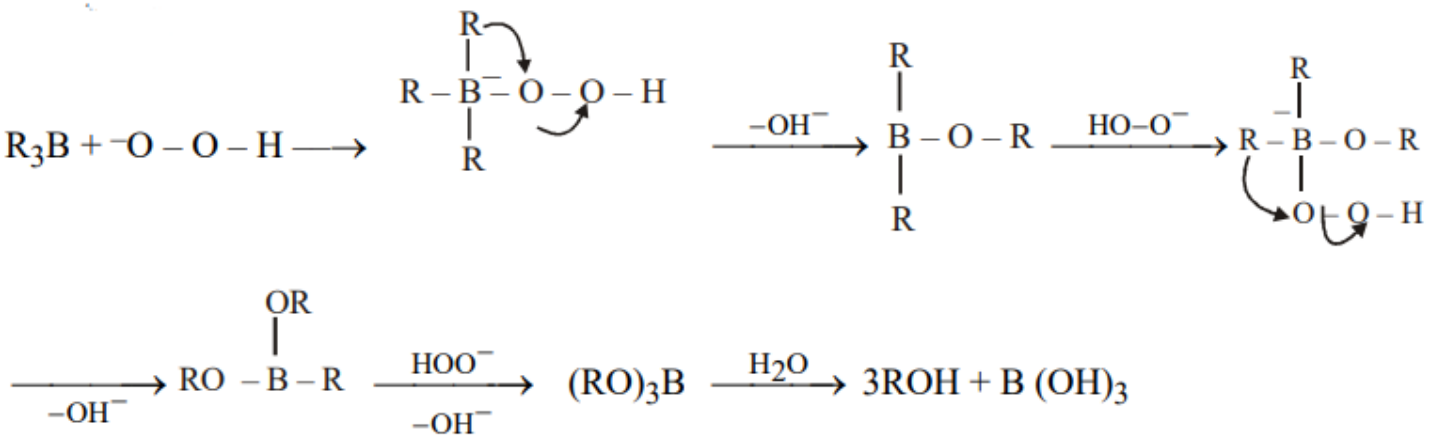
Hydroboration is the process of addition of borane, in which alkyl and alkenyl boranes are perpared. Usually reaction of borane with olefins gives trialkylbornae (R_3B).



Hydrocarbon is regioselective. The addition of $\mathrm{H}-\mathrm{B}$ to alkene follows Markownikoff's addition rule.

Regioselectivity in the hydrocarbon of alkenes with borane is increased by using hindered boranes e.g., disiamyl borane, thexylborane, catechol borane, 9-BBN etc.

Addition of borane to alkene, followed by oxidation gives alcohol (anti-Markownikoff's orientation)

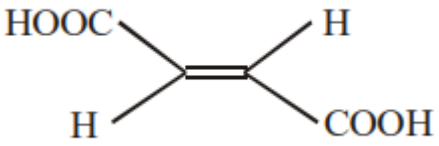


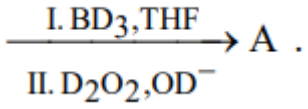
Trialkylborane on treatment with $\mathrm{CH_3COOH}$ or dil. $\mathrm{H_2SO_4}$ gives corresponding alkane and on treatment with

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$AgNO_3, OH^-$ give coupled alkane.

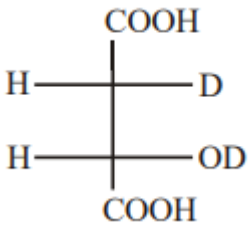
Question:



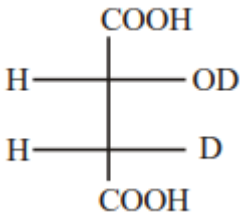


The final product A is

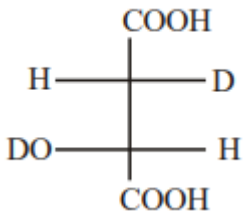
(A)



(B)



(C)

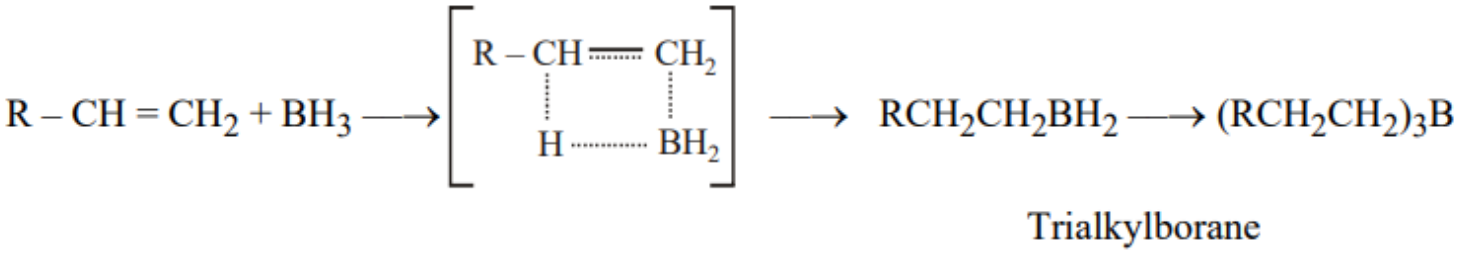


(D) Both A and B

Q.13

Passage:

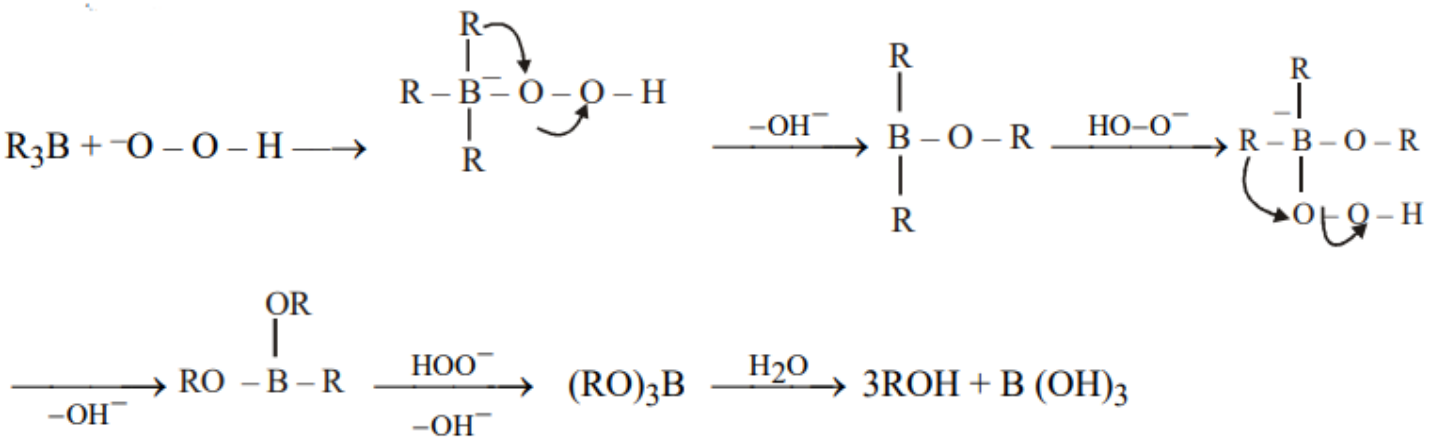
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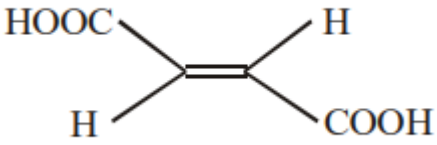
Hydrocarbon is regioselective. The addition of $H - B$ to alkene follows Markownikoff's addition rule. Regioselectivity in the hydrocarbon of alkenes with borane is increased by using hindered boranes e.g., disiamyl borane, thexylborane, catechol borane, 9-BBN etc.

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Addition of borane to alkene, followed by oxidation gives alcohol (anti-Markownikoff's orientation)



Trialkylborane on treatment with CH_3COOH or dil. H_2SO_4 gives corresponding alkane and on treatment with $AgNO_3, OH^-$ give coupled alkane.



Question: $CH_3C \equiv CC_2H_5 \xrightarrow[\text{II. } H_2O_2,OH^-]{\text{I.Sia}_2\text{BH}} L \xrightarrow{\text{Tautomerisation}} M$

Which of the following is in correct regarding above process ?

(A) Reaction with Sia_2BH forms dialkylborane.

(B) L is mixture of diasteromeric products

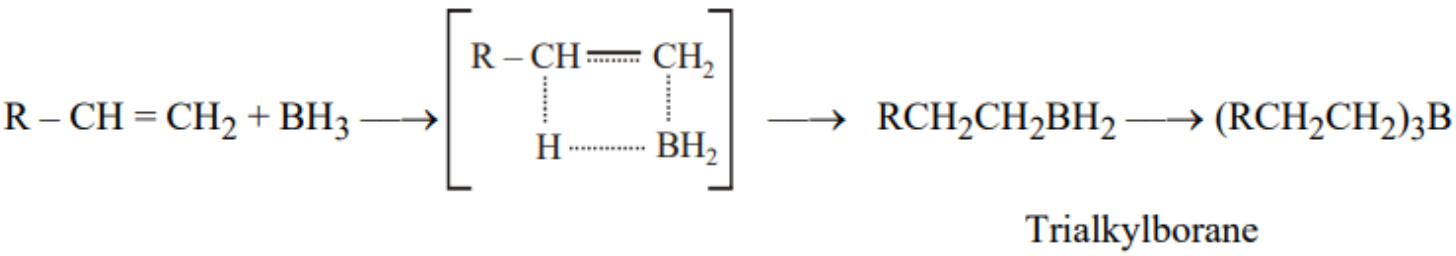
(C) Formation of L is highly regioselective

(D) The product M is acetone

Q.14

Passage:

Hydroboration is the process of addition of borane, in which alkyl and alkenyl boranes are perpared. Usually reaction of borane with olefins gives trialkylbornae (R_3B).

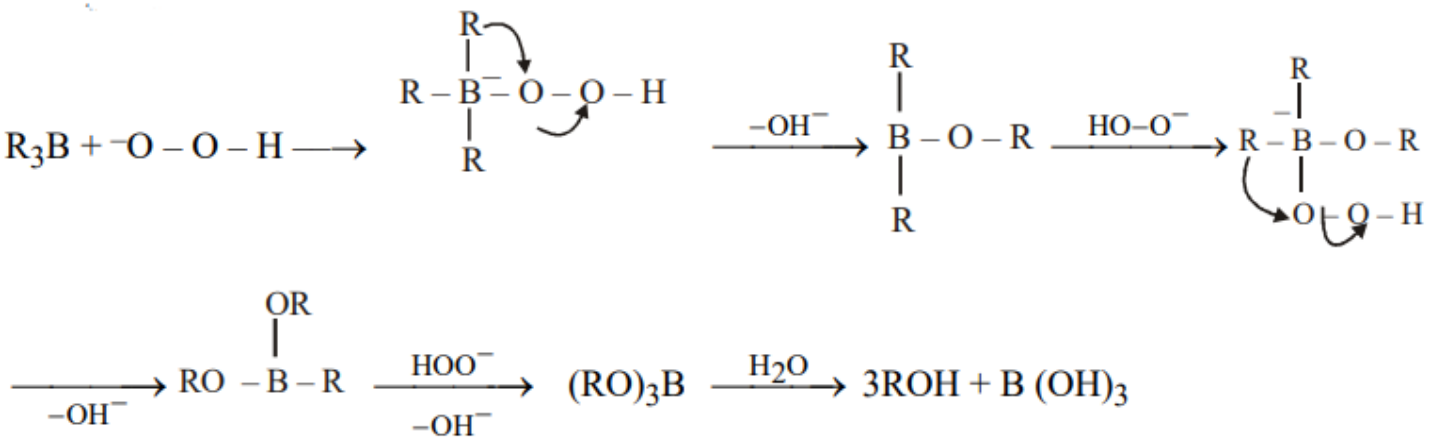


Trialkylborane

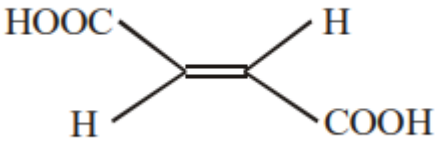
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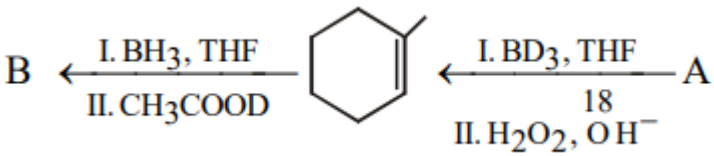
Addition of borane to alkene, followed by oxidation gives alcohol (anti-Markownikoff's orientation)



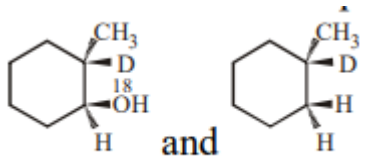
Trialkylborane on treatment with CH_3COOH or dil. H_2SO_4 gives corresponding alkane and on treatment with $AgNO_3, OH^-$ give coupled alkane.

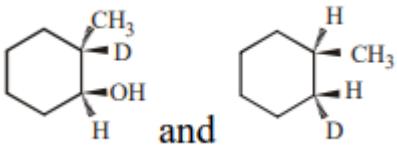


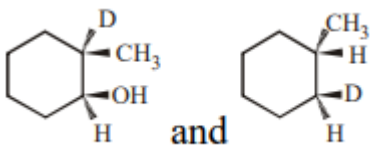
Question:

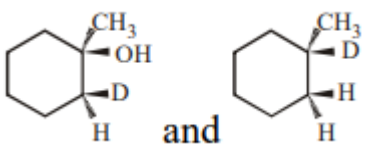


Products A and B are respectively —

(A) 

(B) 

(C) 

(D) 

Q.15

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Passage:

A hydrocarbon P having molecular formula C_3H_4 on treatment with sulphuric acid yields Q having M.F. C_3H_6O. P on treatment with sodium in ether yields sodium salt of hydrocarbon having M.F. $C_3H_3Na(R)$, when Q and R are treated together, an adduct is formed, which on acidicfication yields S having M.F. $C_6H_{10}O$. S on further treatment with H_2 in presence of $Pd - BaSO_4/$ quinoline yields T having M.F. $C_6H_{12}O$, which may exhibits geometrical isomerism. Q on treatment with Mg/ether followed by aqueous work up yields W.

Question: Which of the following statement is correct for the formation of Q from P ?

(A) The formation of Q involves

$$H_2C=CH-\overset{\oplus}{C}H_2$$

(allyl carbocation).

(B) The formation of Q involves the formation of

$$H_3C-\overset{\oplus}{C}=CH_2$$

carbocation as kinetically controlled product.

(C) Q is allyl alcohol.

(D) Q is propanal.

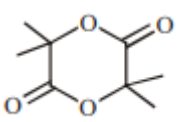
Q.16

Passage:

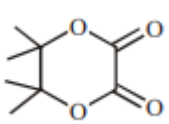
A hydrocarbon P having molecular formula C_3H_4 on treatment with sulphuric acid yields Q having M.F. C_3H_6O. P on treatment with sodium in ether yields sodium salt of hydrocarbon having M.F. $C_3H_3Na(R)$, when Q and R are treated together, an adduct is formed, which on acidicfication yields S having M.F. $C_6H_{10}O$. S on further treatment with H_2 in presence of $Pd - BaSO_4/$ quinoline yields T having M.F. $C_6H_{12}O$, which may exhibits geometrical isomerism. Q on treatment with Mg/ether followed by aqueous work up yields W.

Question: The compound T on oxidative ozonolysis yields acetic acid and a compound having $M.F.C_4H_8O_3$ (U). The new compound U on heating gives -

(A)



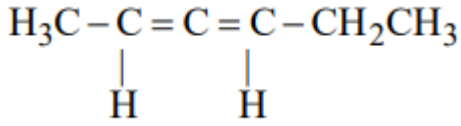
(B)



(C)

$$H_3C-CH=\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}-CH_3$$

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(D) 

Q.17

Passage:

A hydrocarbon P having molecular formula C_3H_4 on treatment with sulphuric acid yields Q having M.F. C_3H_6O. P on treatment with sodium in ether yields sodium salt of hydrocarbon having M.F. $C_3H_3Na(R)$, when Q and R are treated together, an adduct is formed, which on acidicfication yields S having M.F. $C_6H_{10}O$. S on further treatment with H_2 in presence of $Pd-BaSO_4/$ quinoline yields T having M.F. $C_6H_{12}O$, which may exhibits geometrical isomerism. Q on treatment with Mg/ether followed by aqueous work up yields W.

Question: Which of the following statement is correct for the formation of W from Q ?

(A) W is 1,2 -diol and its formation involves carbanion intermediate

(B) W is 1,2 -diol and its formation involves radical intermediate

(C) W is 1,2 -diol and its formation radical-anion intermediate

(D) W is 1,2 -diol and its formation involves carbocation intermediate

Q.18

Match the following :

Column I	**Column II**
(A) Markownikoff's product	(p) $CH_3CH=CHCH_3 \xrightarrow{HBr}$
(B) Peroxide effect	(q) $R_3\overset{+}{N}-CH=CH_2 \xrightarrow{HCl}$
(C) Anti-Markownikoff's product	(r) $CH_3CH=CH_2 \xrightarrow[(CH_3COO)_2]{HCl}$
(D) Mixture of stereoisomers	(s) $CH_3NHCH=CH_2 \xrightarrow{HBr}$
	(t) $CH_3CH=CH_2 \xrightarrow[(CH_3COO)_2]{HBr}$
	(u) $CH_3CH=CHCH_3 \xrightarrow[(CH_3COO)_2]{HBr}$

(A) A-(rs) B-(t) C-(qt) D-(ps)

(B) A-(rs) B-(tu) C-(qt) D-(ps)

(C) A-(rs) B-(u) C-(qt) D-(ps)

(D) A-(rs) B-(ut) C-(t) D-(ps)

Q.19

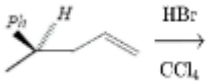
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In solvolysis of 1,2 -dimethyl propyl p-toluene sulphonate in acetic acid at $75^{\circ}C$, how many (alkene + substitution products) will be formed ?

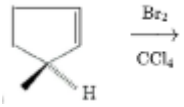
Q.20

How many of the following reactions leads to the formation of diastereomers?

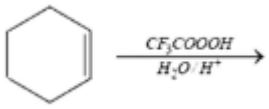
i)



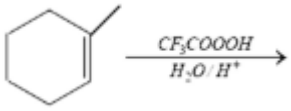
ii)



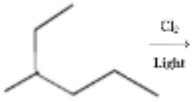
iii)



iv)



v)



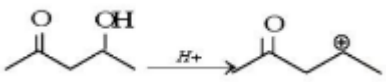
(possible in any monochloro derivative

Answers & Solutions

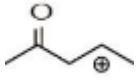
Q.1 Answer:

—

Solution:



Is less stable than

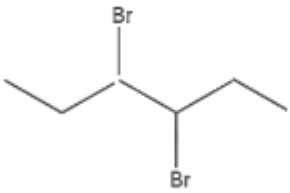


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due to resonance

formed from (1) compound. The other two don't undergo dehydration.

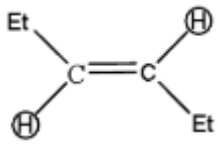
Q.2 Answer:



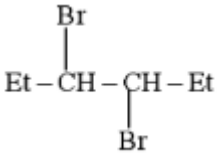
Solution:

(a) Et-C $= C - Et$

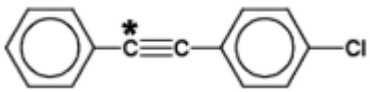
(b)



(c)

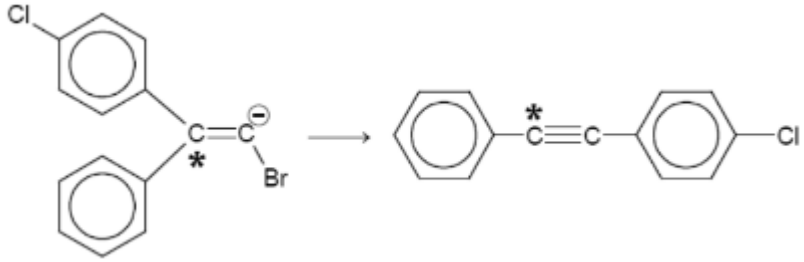


Q.3 Answer:



Solution:

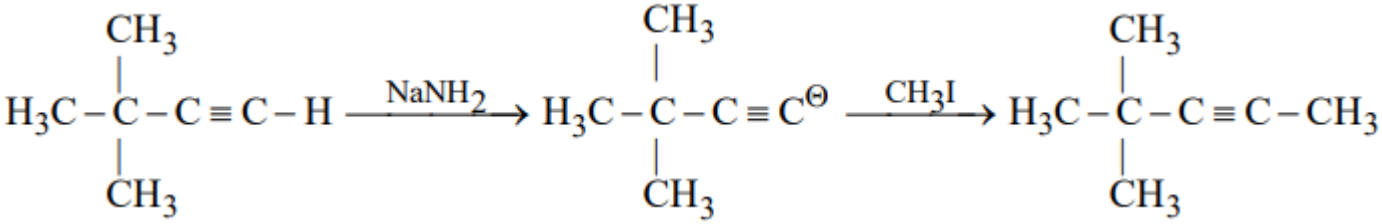
Anti group migrates



Q.4 Answer:

(iii) + (iv)

Solution:

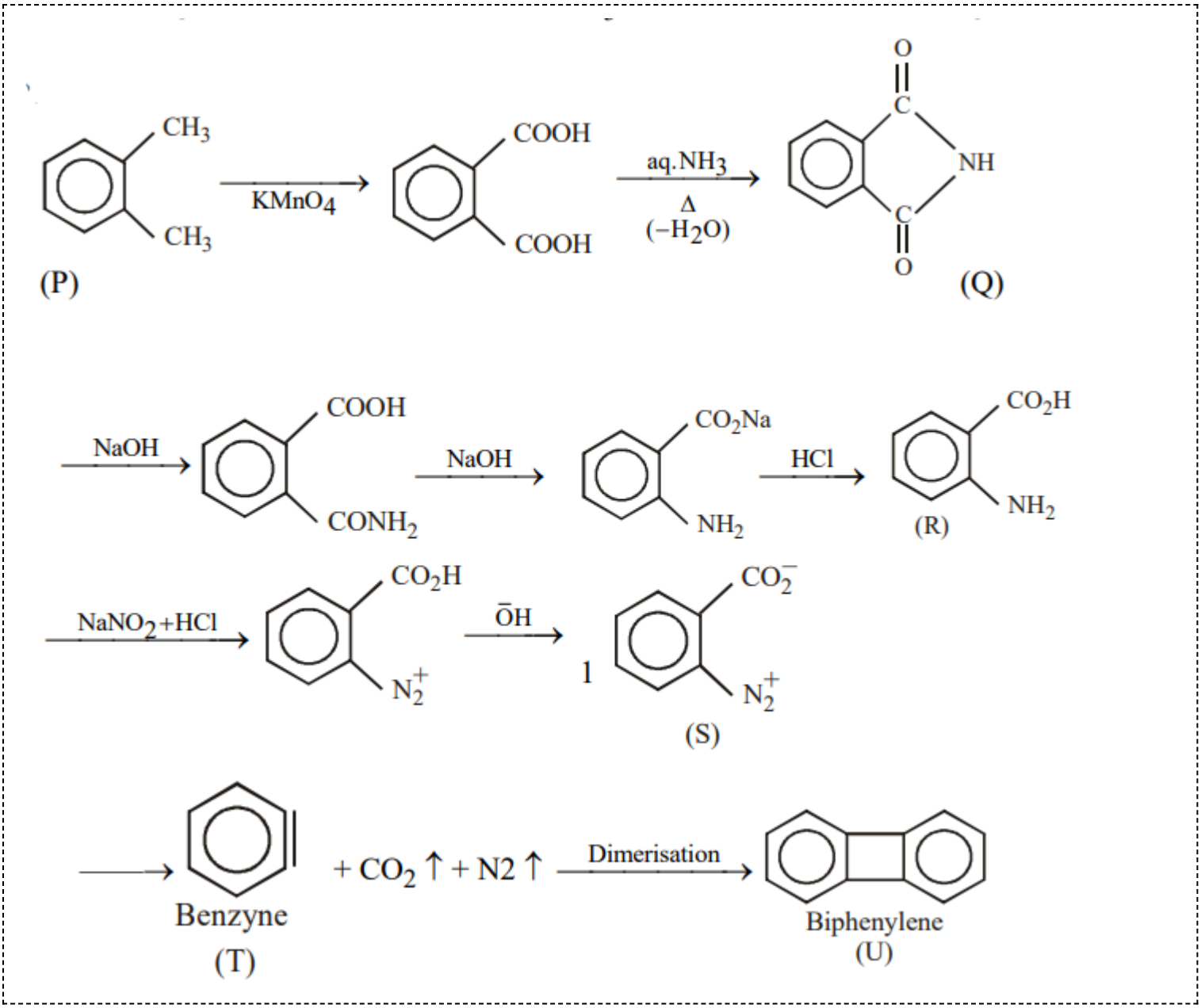


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Q.5 Answer:

9

Solution:

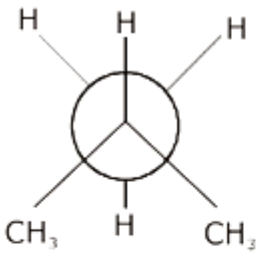


Q.6 Answer:

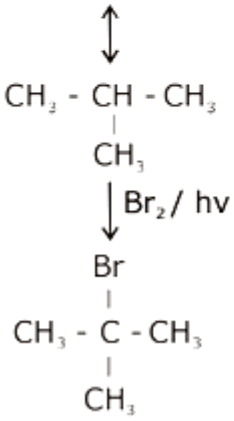
I > III > II

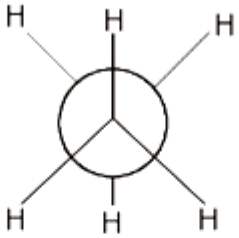
Solution:

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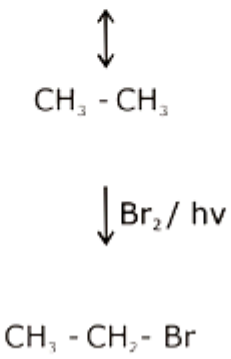


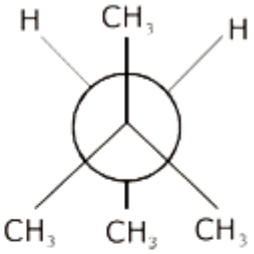
(I)



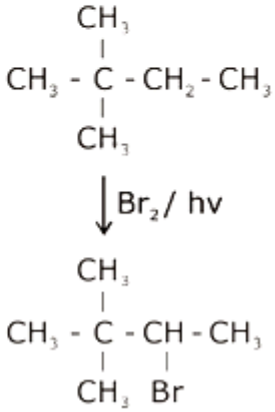


(II)





(III)



[Most stable free radical is formed as intermidiate]

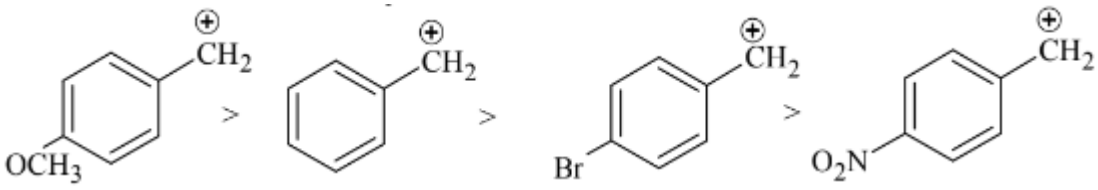
Order of Bromination = I > III > II >

Q.7 Answer:

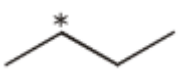
III > I > IV > II

Solution:

Substitution of $-OH$ by Br atom takes place by carbocationicintermediate formation in rds. The stability of carbocation involved is as follows:

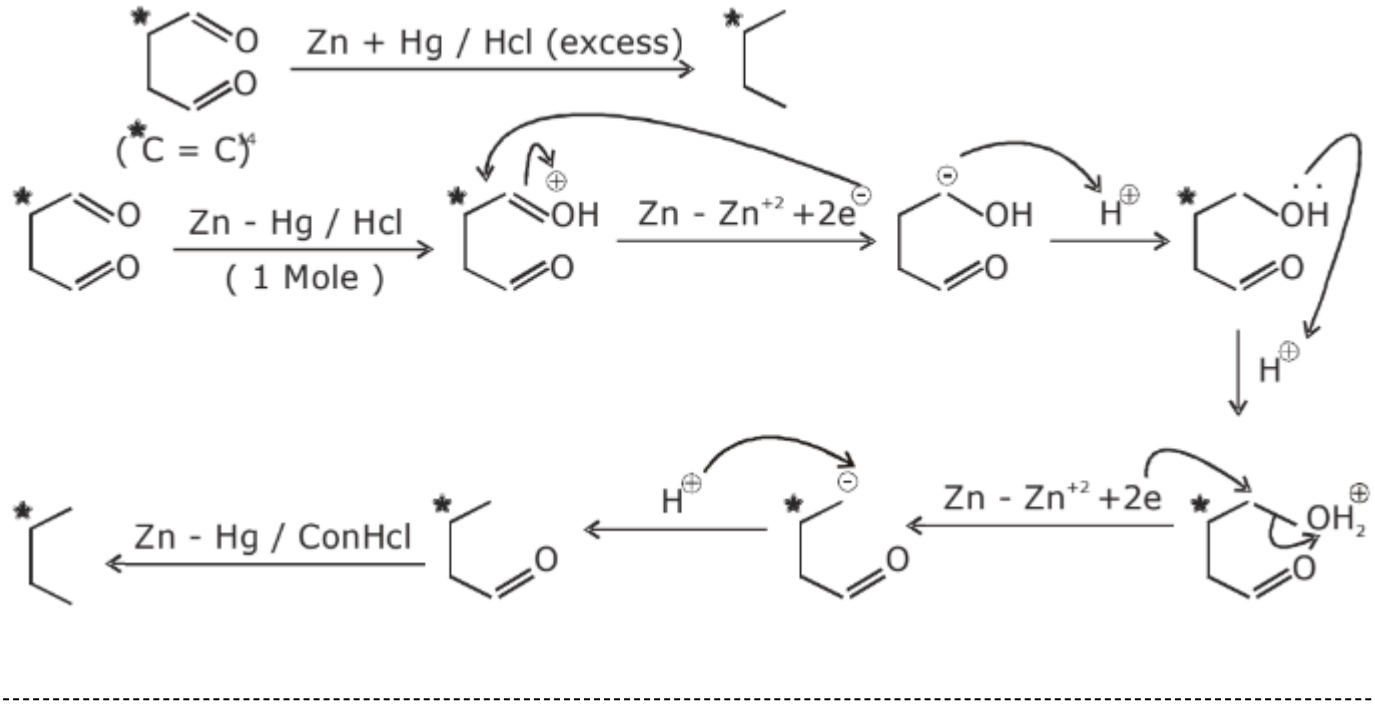


Q.8 Answer:

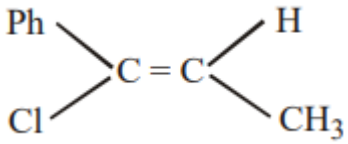


Solution:

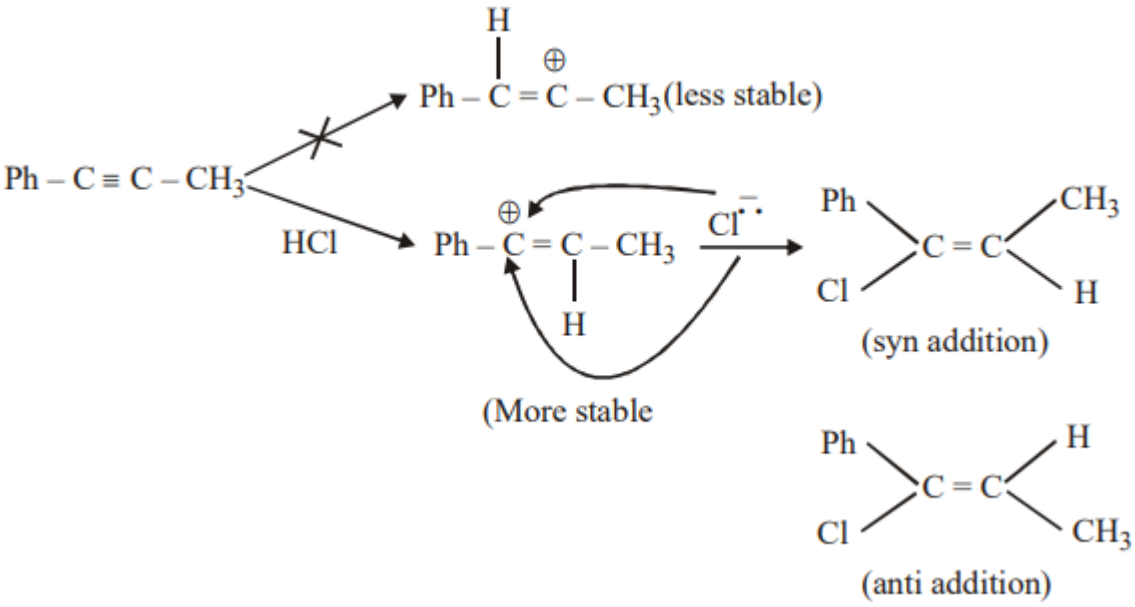
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Q.9 Answer:



Solution:



Q.10 Answer:

Stereospecific reaction

Solution:

Reaction of Byer reagent is : Stereospecific, Stereoselective, Syn addition

Q.11 Answer:

Hydrolysis of phenyl magnesium iodide - Benzene

Solution:

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The huckel rule to account for aromaticity is closed ring of $(4n+2)\pi$ electrons. So option (3) is wrong.

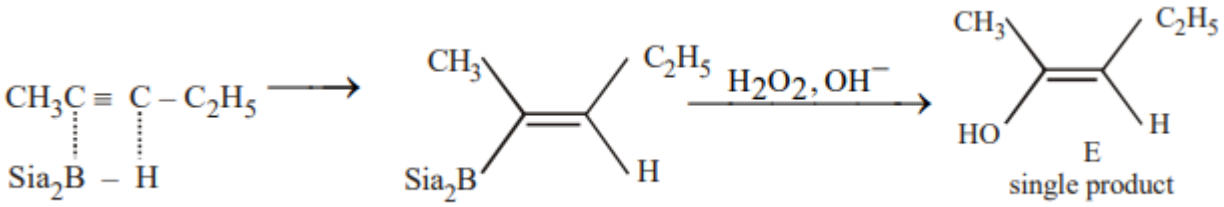
Q.12 Answer:

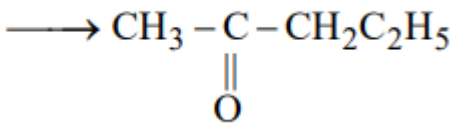
Both A and B

Solution:

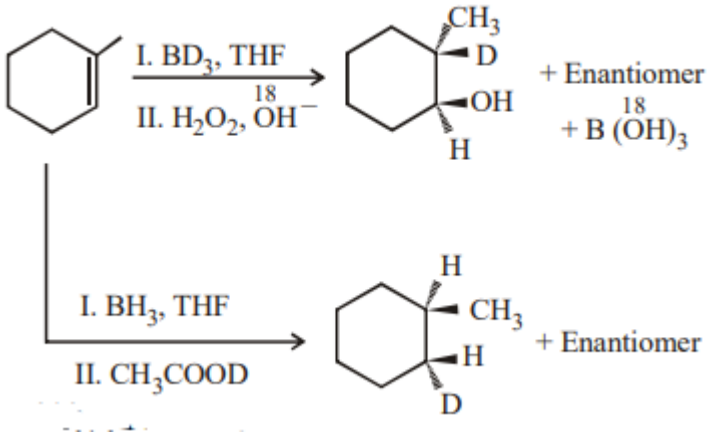
(i) trans + syn $\longrightarrow$ Racemic mixture

(ii)





(iii)



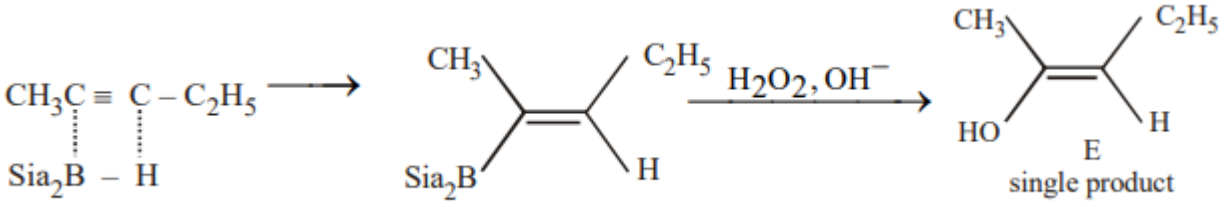
Q.13 Answer:

L is mixture of diasteromeric products

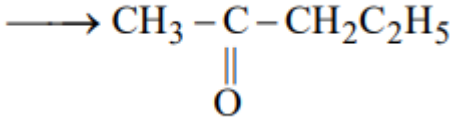
Solution:

(i) trans + syn $\longrightarrow$ Racemic mixture

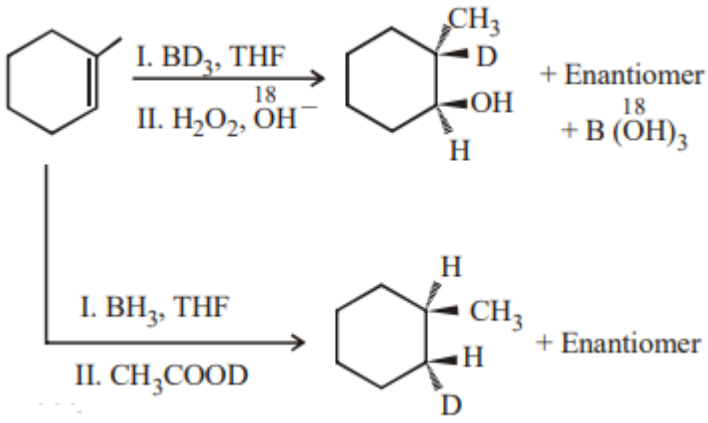
(ii)



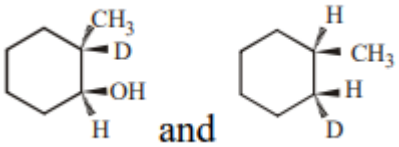
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(iii)



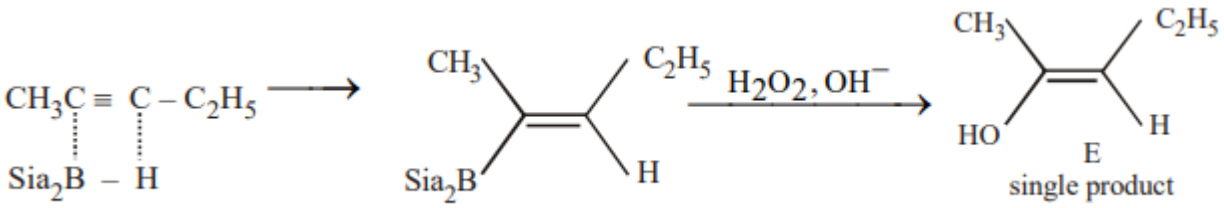
Q.14 Answer:

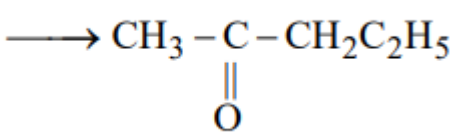


Solution:

(i) trans + syn ⟶ Racemic mixture

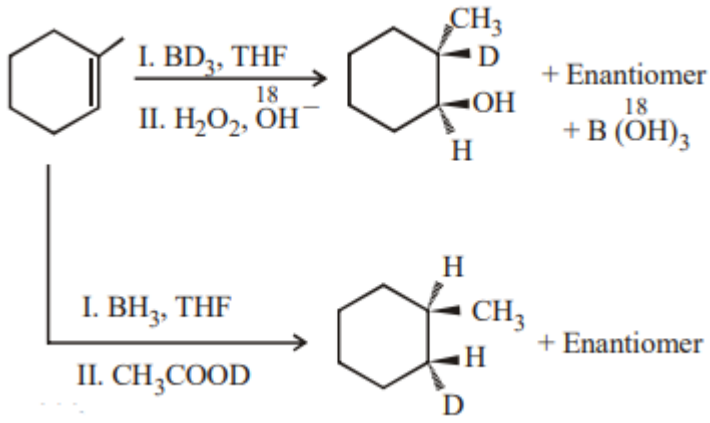
(ii)





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(iii)



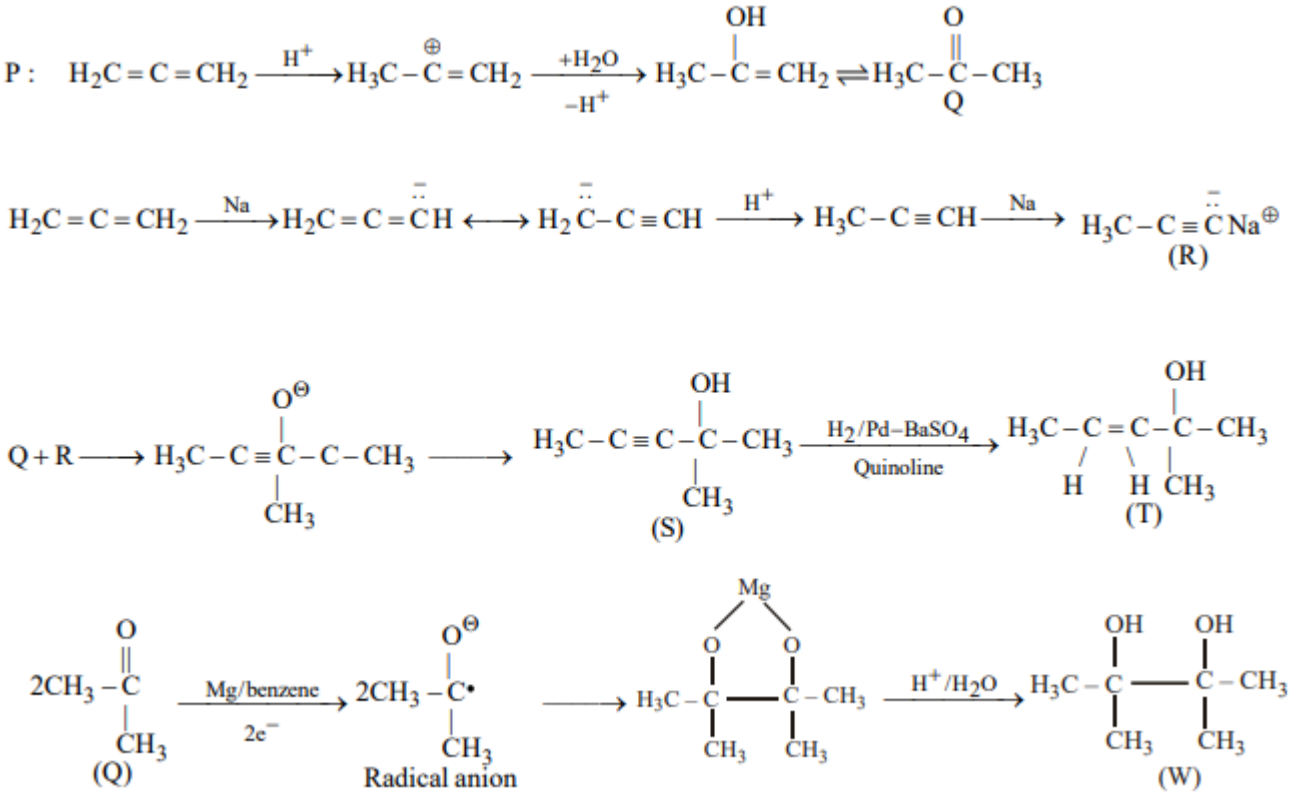
Q.15 Answer:

The formation of Q involves the formation of

$$H_3C-\overset{\oplus}{C}=CH_2$$

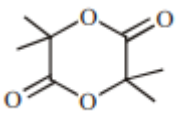
carbocation as kinetically controlled product.

Solution:



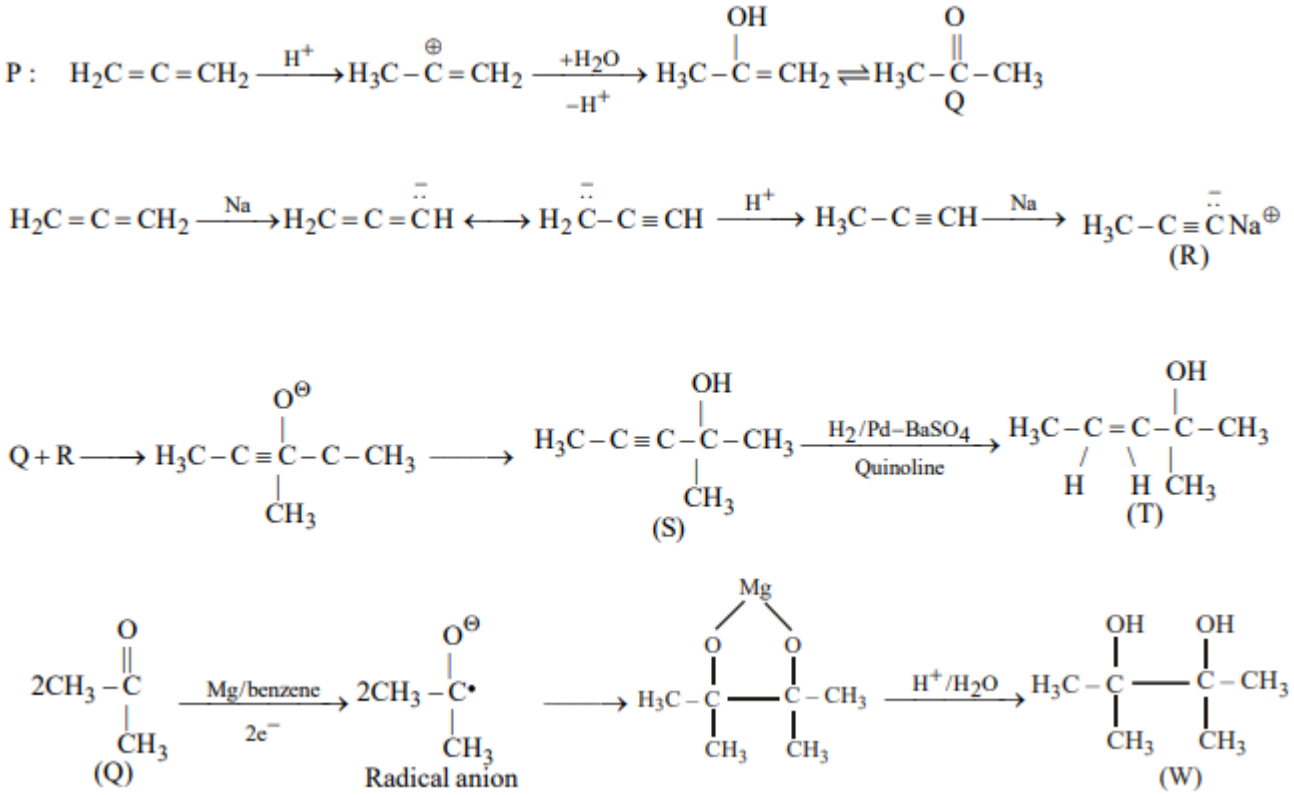
U is a α-hydroxy acid and on heating gives cyclic diester known as lactide.

Q.16 Answer:



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Solution:

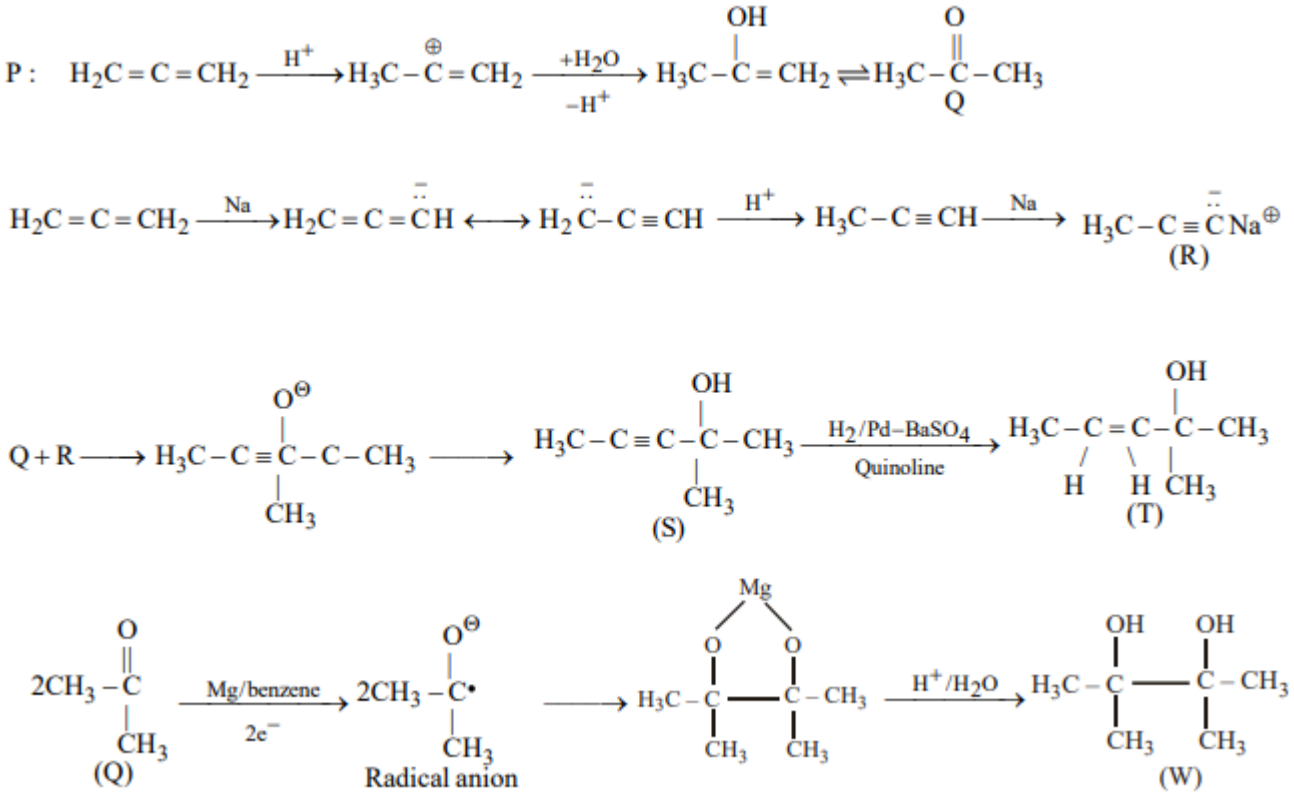


U **is a** α**-hydroxy acid and on heating gives cyclic diester known as lactide.**

Q.17 Answer:

W is 1,2 -diol and its formation radical-anion intermediate

Solution:



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U is a α-hydroxy acid and on heating gives cyclic diester known as lactide.

Q.18 Answer:

A-(rs) B-(tu) C-(qt) D-(ps)

Solution:

$(A)-rs,(B)-tu,(C)-qt,(D)-ps$

$$CH_3CH=CHCH_3 \xrightarrow{HBr} CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2CH_3$$

$$R_3\overset{+}{N}-CH=CH_2 \xrightarrow{HCl} R_3\overset{+}{N}-CH_2-CH_2Cl$$

$$CH_3CH=CH_2 \xrightarrow[(CH_3COO)_2]{HCl} CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-CH_3$$

$$CH_3NHCH=CH_2 \xrightarrow{HBr} CH_3NH-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$$

$$CH_3CH=CH_2 \xrightarrow[(CH_3COO)_2]{HBr} CH_3CH_2CH_2Br$$

$$CH_3CH=CHCH_3 \xrightarrow[(CH_3COO)_2]{HBr} CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2CH_3$$

Q.19 Answer:

—

Solution:

$$CH_3-\underset{\displaystyle OTs}{\underset{|}{CH}}-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3 \xrightarrow[75°C]{AcOH} CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{\oplus}{CH}-CH_3 \xrightarrow{1,2H^- \text{shift}} CH_3-\underset{\displaystyle \oplus}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH_2-CH_3$$

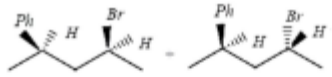
$$\xrightarrow{AcOH} CH_3-\underset{\displaystyle OAc}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH_2-CH_3 + CH_3-\underset{\displaystyle OAc}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{CH}}}}-CH-CH_3 + \text{3-alkene will be obtained.}$$

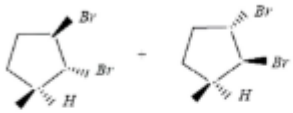
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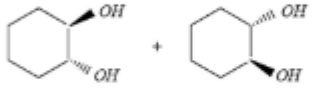
Q.20 Answer:

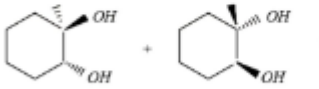
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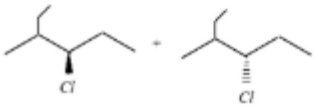
Solution:

(i)  Diastereomers

(ii)  diastereomers

(iii)  enantiomers

(iv)  enantiomers

(v)  diastereomers

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